MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.7: Unit 1 highlights

Big questions. How do vectors represent data, how do matrices act on vectors, and what is special about linear functions?
Notation. $\|\mathbf{v}\|$, $\mathbf{u} \cdot \mathbf{v}$, $\mathrm{cosim}(\mathbf{u}, \mathbf{v})$, $A\mathbf{x}$, $K\mathbf{q}$, $\alpha^T V$, $S = QK^T$, A_{att}, $H = A_{\mathrm{att}} V$, $T(\mathbf{x}) = A\mathbf{x}$, $\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}$.
Concepts. Vector, distance, dot product, cosine similarity, linear combination, convex combination, data vector, word-count vector, token vector, matrix, matrix-vector product, commuting matrices, linear map, affine map.
Results. Dot products connect length, angle, orthogonality, and cosine similarity. The product $A\mathbf{x}$ can be read as row dot products or as a linear combination of columns. The columns of a 2×2 matrix show where $\mathbf{e}_1$ and $\mathbf{e}_2$ go. Matrix multiplication represents composition and is not commutative in general: AB and BA may differ. Some pairs of matrices do commute. Every linear map between Euclidean spaces is represented by a matrix. An affine map $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$ is linear exactly when $\mathbf{b} = \mathbf{0}$.
Main applications. Vector representations of colors, time series, images, features, customers, documents, and tokens; document ranking by cosine similarity; attention as token matrices, dot-product scores, normalized weights, and weighted averages; geometric matrix actions; composition of linear maps; affine layers.
Connections. Unit 2 uses vectors and matrices to describe reachable outputs and forgotten directions through systems. Unit 3 reuses the unit-square visualization as square-grid visualizations and uses affine approximations locally. Least squares in Orthogonality and least squares and gradients in Beyond linear maps: local linearization and first-order

optimization both reuse dot products. Eigenvalues and second-order optimization uses eigenvectors and quadratic forms to study second-order structure.

Study anchors. U1-LO1 and U1-LO2: Vectors. U1-LO3: Matrices. U1-LO4 and U1-LO5: Matrix-vector product and linear maps. U1-LO6: Matrix multiplication and map composition. U1-LO7: Linear and affine maps. U1-LO8: Applications and computation recap and Linked notebook.

Applications and computations readiness checklist. Given a short Unit 1 calculation, code snippet, table, or diagram, I can identify whether `@` is a dot product, matrix-vector product, or matrix product; decide whether a ranking is based on cosine similarity or distance; read $K\mathbf{q}$ as token-query scores; read $\alpha^T V$ as a weighted average; explain the columns of X and Y in $Y = AX$; use the unit-square visualization for a 2×2 matrix; check dimensions before multiplying; compare AB and BA to determine whether two small matrices commute; and decide whether $\mathbf{x} \mapsto W\mathbf{x} + \mathbf{b}$ is linear or affine.

Common mistakes. Confusing dot product with distance, forgetting to normalize cosine similarity, mixing up rows and columns, multiplying matrices in the wrong order, assuming $AB = BA$, assuming that two non-diagonal matrices can never commute, or calling an affine map linear when the bias is nonzero.